

Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties
Monday and Thursday afternoon sessions; Lectures 11–15

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Monday afternoon — material available: Lectures 1–11

Exercise 1. Global functions as morphisms to the affine line

★ **CORE**

Let X be a scheme.

- (i) Explain why a global section $s \in \Gamma(X, \mathcal{O}_X)$ determines a morphism

$$f_s : X \longrightarrow \mathbb{A}_{\mathbb{Z}}^1.$$

- (ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}_{\mathbb{Z}}^1) \simeq \Gamma(X, \mathcal{O}_X).$$

- (iii) If $X = \mathrm{Spec}(A)$, describe the morphism associated with $a \in A$.

Hint 1 (conceptual). Use the affine-target adjunction for $\mathbb{A}_{\mathbb{Z}}^1 = \mathrm{Spec}(\mathbb{Z}[t])$.

Hint 2 (technical). A unital ring homomorphism $\mathbb{Z}[t] \rightarrow \Gamma(X, \mathcal{O}_X)$ is uniquely determined by the image of t .

Exercise 2. The reduced point and the double point

★ **CORE**

Inside $\mathbb{A}_{\mathbb{k}}^1 = \mathrm{Spec}(\mathbb{k}[t])$, consider

$$P = \mathrm{Spec}(\mathbb{k}[t]/(t)), \quad D = \mathrm{Spec}(\mathbb{k}[t]/(t^2)).$$

- (i) Show that the two closed immersions have the same image as subsets of $\mathbb{A}_{\mathbb{k}}^1$.
(ii) Show that P and D are not isomorphic as schemes.
(iii) Compute the maps on local rings at their unique points.

Hint 1 (conceptual). The two images as subsets are $V(t)$ and $V(t^2)$. Compare their radicals.

Hint 2 (technical). Each source has only one point, so its local ring there is its whole coordinate ring. Reducedness already distinguishes the two schemes.

Exercise 3. A morphism between affine schemes

★ **CORE**

Let

$$\varphi : \mathbb{k}[u, v] \longrightarrow \mathbb{k}[t], \quad u \longmapsto t, \quad v \longmapsto t^2.$$

- (i) Describe the induced morphism $f : \mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^2$.
(ii) Determine $\ker(\varphi)$ and factor f through a closed subscheme of $\mathbb{A}_{\mathbb{k}}^2$.
(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

Hint 1 (technical). The relation $v - u^2$ is in the kernel. Divide an arbitrary $F(u, v)$ by the monic polynomial $v - u^2$, viewing it as a polynomial in v .

Hint 2 (technical). After passing to $\mathbb{k}[u, v]/(v - u^2)$, write down an inverse to the induced map by specifying the image of t .

Exercise 4. An arithmetic spectrum

★★ **CHOICE**

Let $A = \mathbb{Z}[s]/(s^2 - 3)$. Describe the inverse image in $\mathrm{Spec}(A)$ of the points (2) , (3) , and (p) for $p \neq 2, 3$, according to the factorisation of $s^2 - 3$ modulo p .

Hint 1 (technical). The inverse image of (p) is the spectrum of $\mathbb{F}_p[s]/(s^2 - 3)$. Factor the polynomial in the three cases $p = 2$, $p = 3$, and $p \neq 2, 3$.

Thursday afternoon — material available: Lectures 1–14

Exercise 5. Dual numbers and tangent vectors

★★ CORE

Let $D = \text{Spec}(\mathbb{k}[\varepsilon]/(\varepsilon^2))$ and let $p = (a, b) \in \mathbb{A}_{\mathbb{k}}^2(\mathbb{k})$.

(i) Show that morphisms $D \rightarrow \mathbb{A}_{\mathbb{k}}^2$ whose closed point maps to p are uniquely of the form

$$x \mapsto a + u\varepsilon, \quad y \mapsto b + v\varepsilon.$$

(ii) Let $C = \mathbb{W}(f) \subset \mathbb{A}_{\mathbb{k}}^2$. Show that such a morphism factors through C precisely when

$$f(a, b) = 0, \quad f_x(a, b)u + f_y(a, b)v = 0.$$

(iii) Apply this to the parabola $y = x^2$.

Suggested strategy. Translate the morphism into a homomorphism of coordinate rings, then expand f after substituting $a + u\varepsilon$ and $b + v\varepsilon$ and discard every term containing ε^2 .

Hint 1 (technical). Every element of $\mathbb{k}[\varepsilon]/(\varepsilon^2)$ is uniquely $c + d\varepsilon$. Mapping the closed point to (a, b) fixes the constant terms of the images of x and y .

Hint 2 (technical). Prove the first-order formula

$$f(a + u\varepsilon, b + v\varepsilon) = f(a, b) + (f_x(a, b)u + f_y(a, b)v)\varepsilon.$$

Exercise 6. The line with two origins

★★ CORE

Glue two copies

$$X_1 = \text{Spec}(\mathbb{k}[t_1]), \quad X_2 = \text{Spec}(\mathbb{k}[t_2])$$

along $D(t_1) \simeq D(t_2)$ by $t_1 \mapsto t_2$. Let X be the resulting scheme.

(i) Describe the underlying topological space.

(ii) Show that $\Gamma(X, \mathcal{O}_X) \simeq \mathbb{k}[t]$.

(iii) Show that the canonical map $X \rightarrow \text{Spec } \Gamma(X, \mathcal{O}_X)$ identifies the two origins.

(iv) Deduce that X is not affine.

Suggested strategy. Describe the gluing topologically first. Then compute global functions as compatible pairs on the two affine charts. Finally compare the scheme with the spectrum of that global ring.

Hint 1 (technical). A global function is a pair $(f_1, f_2) \in \mathbb{k}[t_1] \times \mathbb{k}[t_2]$ whose images agree in $\mathbb{k}[t, t^{-1}]$. Use the injectivity of $\mathbb{k}[t] \rightarrow \mathbb{k}[t, t^{-1}]$.

Hint 2 (conceptual). The canonical map to $\text{Spec } \Gamma(X, \mathcal{O}_X)$ restricts to the ordinary coordinate map on both charts. Track the two origins under this map.

Exercise 7. Global functions on the projective line

★★ CHOICE

Construct $\mathbb{P}_{\mathbb{k}}^1$ by gluing $U_0 = \text{Spec}(\mathbb{k}[t])$ and $U_1 = \text{Spec}(\mathbb{k}[s])$ along $s = t^{-1}$. Prove that

$$\Gamma(\mathbb{P}_{\mathbb{k}}^1, \mathcal{O}_{\mathbb{P}^1}) = \mathbb{k}.$$

Hint 1 (technical). A global function is a pair $f(t) \in \mathbb{k}[t]$, $g(s) \in \mathbb{k}[s]$ satisfying $f(t) = g(t^{-1})$ in $\mathbb{k}[t, t^{-1}]$. Compare the possible positive and negative powers of t .

Exercise 8. The projective conic from its standard opens

★★ CORE

Let

$$S = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \quad C = \text{Proj}(S).$$

(i) Compute $(S_{x_0})_0$ and $(S_{x_2})_0$.

(ii) Identify $D_+(x_0)$ and $D_+(x_2)$ with affine lines.

(iii) Describe their overlap and recover the gluing rule for $\mathbb{P}^1$.

Suggested strategy. Compute the degree-zero parts of the two homogeneous localisations, then express the transition function on the overlap.

Hint 1 (technical). On $D_+(x_0)$ use $t = x_1/x_0$ and the relation $x_2/x_0 = t^2$. Do the symmetric computation on $D_+(x_2)$.

Hint 2 (technical). On the overlap, both x_0 and x_2 are non-zero. Compare the two affine parameters using $x_0x_2 = x_1^2$.

Exercise 9. Connectedness and idempotents

★★ **CORE**

- (i) Prove that $\text{Spec}(A)$ is disconnected if and only if A has a non-trivial idempotent.
- (ii) Apply this to $A = \mathbb{k}[t]/(t^2 - t)$.
- (iii) Show that $\text{Spec}(\mathbb{k}[x, y]/(xy))$ is connected although reducible.

Hint 1 (conceptual). A decomposition of $\text{Spec}(A)$ into two disjoint open-and-closed subsets should correspond to a function taking the values 0 and 1 on the two pieces.

Hint 2 (technical). Starting from an idempotent e , use the two principal opens $D(e)$ and $D(1 - e)$. Conversely, translate a clopen decomposition into a product decomposition of A .

Hint 3 (technical). For $\mathbb{k}[x, y]/(xy)$, an idempotent restricts to idempotents on both axes. Compare their values at the common origin.

Exercise 10. Reduction and its universal property

★★ **CORE**

Let

$$A = \mathbb{k}[x, y]/(x^2, xy), \quad X = \text{Spec}(A).$$

- (i) Compute the nilradical and X_{red} .
- (ii) Let R be a reduced $\mathbb{k}$ -algebra. Show that every homomorphism $A \rightarrow R$ factors uniquely through A_{red} .
- (iii) Translate this into a statement about morphisms of schemes.

Hint 1 (conceptual). Find the nilradical first. If R is reduced, the image in R of every nilpotent element of A must vanish.

Hint 2 (technical). Use this to show that the kernel of any homomorphism $A \rightarrow R$ contains the nilradical. The desired factorisation is then the universal property of a quotient ring.

Hint 3 (conceptual). Remember to reverse arrows when translating the factorisation into a statement about schemes.

Exercise 11. A weighted projective chart

★★★ **OPTIONAL**

Let $R = \mathbb{k}[x, y, z]$ with $\deg x = \deg y = 1$ and $\deg z = 2$, and set $X = \text{Proj}(R)$. Show that $D_+(x)$ and $D_+(y)$ are affine planes and that

$$D_+(z) \simeq \text{Spec}(\mathbb{k}[a, b, c]/(ac - b^2)).$$

Identify the singular point of this chart.

Suggested strategy. For each standard open $D_+(f)$, compute the degree-zero subring $(R_f)_0$. The chart at z is the only one where the grading produces a relation.

Hint 1 (technical). On $D_+(z)$ consider the degree-zero elements

$$a = x^2/z, \quad b = xy/z, \quad c = y^2/z.$$

They satisfy one evident relation.

Hint 2 (technical). To locate the singular point of $ac - b^2 = 0$, use the partial derivatives with respect to a, b, c .

After Friday's lecture or preparation for Week 4

Exercise 12. Components and generic points

★★ **CHOICE**

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

Show that $I = (x) \cap (y, z)$. Determine the irreducible components, their generic points and dimensions, and decide whether X is connected or integral.

Hint 1 (conceptual). For an affine scheme, irreducible components correspond to minimal prime ideals, and their generic points are those minimal primes viewed as points of the spectrum.

Hint 2 (technical). To prove $(x) \cap (y, z) = (xy, xz)$, write an element of the intersection as xg and reduce modulo (y, z) .

Hint 3 (conceptual). Use the definition of irreducibility and inspect whether the two components meet. Reducedness comes from the expression of I as an intersection of prime ideals.

Exercise 13. The function field of the cusp

★★ CHOICE

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral and that $K(C) \simeq \mathbb{k}(t)$.

Hint 1 (technical). Use the map $x \mapsto t^2, y \mapsto t^3$. Every class in the coordinate ring has a representative $A(x) + yB(x)$; parity of exponents proves injectivity.

Hint 2 (technical). The fraction field is contained in $\mathbb{k}(t)$. Recover t as a quotient of the images of x and y .